

Mathematics 8 BSH877

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of BSH&M (Mathematics) In-Semester Examination II, Academic Year: 2025-26 (Semester – II)	
Class: SY B.Tech.	Subject: M-IV (Statistical methods and Complex analysis).	Subject Code: BSC275
Date: 25/03/2026	Time: 14:30 to 15:15pm.	Max. Marks: 20

Note: i) Assume suitable data whenever necessary and state it clearly.
 ii) Nonprogrammable calculators are allowed.

Q.1	Attempt any four			
a	i) A bag contains 8 white balls and 6 black balls. If 5 balls are drawn at random without replacement, find the probability that exactly 3 of the selected balls are white and 2 are black. ii) An unbiased die is thrown. What is the mathematical expectation of the number of points?	05	CO2	BT2
b	Three identical boxes, I, II, and III, contain 4 white & 3 red balls; 3 white & 7 red balls; and 2 white & 3 red balls, respectively. A box is chosen at random, and a ball is drawn from it. If the ball is found to be white, what is the probability that Box II was selected?	05	CO2	BT2
c	The Probability density function (p. d. f.) of a continuous random variable X is $f(x) = \frac{1}{2} - ax$; ($0 \leq x \leq 4$). Determine, i) The value of the constant a . ii) The cumulative distribution function (c. d. f.). iii) The probability $P\left(\frac{3}{2} \leq X \leq \frac{5}{2}\right)$.	05	CO2	BT3
d	i) The Probability that an entering college student will be a graduate is 0.4. Determine the probability that out of 5 entering students at least 3 will be graduate. ii) A coin is tossed 10 times. Find the probability of getting exactly 6 heads.	05	CO2	BT2
e	A box contains 4 white and 6 black balls. If 3 balls are drawn at random, Find the mathematical expectation of the number of white balls.	05	CO2	BT3

[BT1: Remembering, BT2: Understanding, BT3: Applying, BT4: Analysing, BT5: Evaluating, BT6: Creating]

- CO1. To develop techniques of data interpretation.
- CO2. Develop problem solving techniques needed to accurately calculate probabilities and describe the properties of discrete and continuous distribution functions.
- CO3. Use statistical tests in testing hypotheses on data.
- CO4. Determine whether a given function is analytic and apply analyticity on harmonic functions. and conjugates of harmonic functions.
- CO5. Transform a region to another region using conformal mapping and will be able to evaluate complex integrals, classify singularities and poles, find residues.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Vishnupuri, Nanded- 431606.

Exam: Inter-Sem Examination - II (SEM-II 2025-26)

Class: S. Y. B. Tech (CSE, INFT) Department: Mathematics

Course Name and Code: M-IV Vector Calculus and Statistical Methods, BSC-276

Day and Date: Wednesday, 25/03/2026 Time: 14:30 - 15:15 Maximum Marks: 20

- All questions are compulsory.
- Calculators are allowed.

Q 1) Solve any four of the following. (20 marks) [CO3]

1. A and B play a game in which their chances of winning are in the ratio 3:2. Find A's chance of winning at least three games out of the five games played. [BT3, BT5]
 2. Define Binomial distribution. Find the mean and variance of Binomial distribution. [BT1, BT3]
 3. Let X be a continuous normal variable with mean 30 and standard deviation 5. Find the probabilities that (a) $26 \leq X \leq 40$, (b) $X \geq 45$, and (c) $|X - 30| > 5$. Given that for normal standard distribution of Z , $P(-2 \leq Z \leq 2) = 0.8544$, $P(0 \leq Z \leq 0.8) = 0.2881$, $P(0 \leq Z \leq 3) = 0.49865$, $P(0 \leq Z \leq 1) = 0.3413$. [BT3, BT5]
 4. A fair coin is tossed 400 times. Using normal approximation, find the probability of obtaining (i) exactly 200 heads; (ii) less than 210 heads; (iii) between 190 and 210 heads, both inclusive. Given that the area under the standard normal curve: between $z = 0$ and $z = 0.05$ is 0.0199, between $z = 0$ and $z = 0.95$ is 0.3289, between $z = 0$ and $z = 1.05$ is 0.3531. [BT3, BT5]
 5. In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of errors per page, find the probability that a random sample of 7 pages will contain no error. [BT3, BT5]
-

Bloom's Taxonomy:

BT1: Remembering	BT2: Understanding	BT3: Applying
BT4: Analyzing	BT5: Evaluating	BT6: Creating

Course Outcomes:

- CO1: Evaluate line integrals, surface integrals, and volume integrals and convert line integrals into area integrals and surface integrals into volume integrals using integral theorems.
- CO2: To develop techniques of data interpretation.
- CO3: Develop problem solving techniques needed to accurately calculate probabilities and describe the properties of discrete and continuous distribution functions.
- CO4: Use statistical tests in testing hypotheses on data.
- CO5: Compute covariances, and correlations, Apply the tests of goodness of fit.



**SGGS Institute of Engineering and Technology, Vishnupuri,
Nanded-431606.**

Department of BSH&M(Mathematics)

Academic Year: 2025-26 (Semester-II)

In-Semester Examination-II

Class: S.Y. B. Tech.	Paper: Mathematics-IV (Statistical and Numerical Methods)	Course Code: BSC274
Date: 25/03/2026	Time: 14:30-15:15	Max. Marks: 20

Note: (i) Attempt all questions. (ii) Assume suitable data if necessary (iii) Calculator is allowed.

Q1	Attempt any four.	Marks	CO	BT																
(a)	There are three children in a family. Find the probability that all the children are boys, (i) if no prior information is available about the children, (ii) if it is known that the two eldest are boys, (iii) if it is known that at least two of them are boys.	05	CO1	BT2																
(b)	The probability density function of a continuous random variable is $f(x) = k(x - 1)(2 - x)$; $(1 \leq x \leq 2)$ Determine (i) the value of the constant k , (ii) the cumulative distribution function.	05	CO2	BT3																
(c)	The mean and variance of a Binomial distribution are 4 and $4/3$. Find the distribution and the probability of at least one success.	05	CO2	BT2																
(d)	A discrete random variable X has the following probability mass function <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>$p(x)$</td><td>k</td><td>3k</td><td>5k</td><td>7k</td><td>9k</td><td>11k</td><td>13k</td></tr></table> (i) Find the value of k. (ii) Find $P(X < 4)$, $P(X \geq 5)$ & $P(3 < X \leq 6)$.	x	0	1	2	3	4	5	6	$p(x)$	k	3k	5k	7k	9k	11k	13k	05	CO2	BT2
x	0	1	2	3	4	5	6													
$p(x)$	k	3k	5k	7k	9k	11k	13k													
(e)	A random variable X follows Poisson distribution with parameter 3, Find the probabilities that the variable assumes the values (i) less than 3 (ii) at least 2 (Given that $e^{-3} = 0.0498$).	05	CO2	BT3																

[BT1: Remembering, BT2: Understanding, BT3: Applying, BT4: Analysing, BT5: Evaluating, BT6: Creating]

CO1. Develop the skills of Laplace Transforms, Fourier series and Fourier Transforms and their inverses.

CO2. Develop the skills of solving Partial differential equations.

CO3. Solve ODE's and PDE's using the properties of Laplace transform, Fourier series and Fourier Transforms.

CO4. Determine solutions of PDE for vibrating string and heat conduction.

CO5. Evaluate line integrals, surface integrals, and volume integrals and convert line integrals into area integrals and surface integrals into volume integrals using integral theorems.



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Nanded**
Department of BSH&M (Mathematics)
In-Semester Examination II,
Academic Year: 2025-26 (Semester – II)

Class: SY B.Tech.	Subject: Indian Constitution	Subject Code: MAC277
Date: 24.03.2026	Time: 14:30 to 15:15	Max. Marks: 20

Note: i) Assume suitable data whenever necessary and state it clearly.

ii) Mobile phones and calculator not allowed

Q.N.				
1	Explain the Constitutional framework for financial relations between the Centre and States	10	CO1	BT4
2	Describe the procedure for amending the Constitution under Article 368	10	CO2	BT2
3	What is the difference between the President rule and the National Emergency	10	CO3	BT1
4	What are some of the most significant Constitutional amendments in Indian history and why are they considered important?	10	CO6	BT2
5	Write short notes on (any two) A) Importance of Directive principles of State policy B) Subjects of Concurrent list C) Federal structure of Indian Constitution	10	CO4	BT4

----ALL THE BEST----

Bloom's Taxonomy: [BT1: Remember BT2: Understand, BT3: Apply BT4: Analyzes, BT5: Evaluate, BT6: Create]

Course Outcomes

CO1. Student will be able to understand how the legislative and financial powers have been distributed between centre and state respectively

CO2. Student will be able to understand the procedure of amendment under Indian constitution under Article 368.

CO3. Student will be able to understand the status and power of president under federal structure of Indian constitution

CO4. Student will be able to understand concept of federalism.

CO5. Student will be able to understand amendment of the constitutional powers and procedure

CO6. Student will be able to understand the historical perspectives of the constitutional amendments in India



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of BSHM

Academic Year: 2025-26 (semester II)

In-Semester Examination II

Class: SY

Course Name: Ethics, Integrity & Aptitude

Course Code: OE-IS-PC202

Date: 30.03.2026

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Answer ANY 2 of the following questions. (100 Words)	Marks	CO	BT
Q.1	a	What is Aptitude? How is it different from Attitude?	5	CO1	BT1
	b	Organise the differences between 'Sympathy', 'Empathy' and 'Compassion' as foundational values for the civil services.	5	CO1	BT3
	c	Do you believe 'Integrity' is the most important value for the civil servants in performance of their duty? Analyse.	5	CO2	BT4
	d	Compare and contrast 'Impartiality' and 'Non-Partisanship' with respective definitions and dimensions.	5	CO4	BT2
		Answer ANY 1 of the following questions. (150 Words)			
Q.3		Define 'Emotional Intelligence (EI)'. Discuss the attributes and role of EI in administration and governance.	10	CO3	BT6
Q.4		Evaluate the key ideas of any 2 'Moral Thinkers & Philosophers' from India (Any 1) and the World (Any 1).	10	CO5	BT5

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To grasp fundamental concepts of Ethics, Integrity and Aptitude.
CO 2	To recognize the significance of Ethics and Integrity, and its impact on daily life.
CO 3	To utilize contextual knowledge to evaluate and address administrative issues.
CO 4	To relate course topics to contemporary national and societal issues, enhancing their effectiveness in multidisciplinary settings.
CO 5	To develop the skills and mindset necessary for independent and lifelong learning.

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded		
	Class: Second Year	Academic Year: 2025-26	Semester: II
	Examination: ISE II	Subject: Business Communication for Engineers	Course Code: OE-IS-EN202
	Date: 30/03/2026	Time: 14:30-15:15	Max. Marks :20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q. No.	Attempt any <u>two</u> of the following questions.	Marks	CO	BT
Q.1	What is proofreading? Explain it with at least ten symbols and its examples.	10	CO2	BT2
Q.2	Describe public speaking techniques for influential speech.	10	CO3	BT3
Q.3	With the help of suitable examples, explain the process of handling questions and answers for effective presentation skills.	10	CO3	BT4

== ALL THE BEST ==



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
ISE-II Examination (Semester - II)
Academic Year: 2025-26

Class: SY-B.Tech.
(CE/CSE/IT/CH/TEXT)

Subject: Project Management

Subject Code: HSSM-201

Date: 25-03-2026

Time: 14:30 pm to 15:15 pm

Max. Marks: 20

Note:

- i. Figures to right indicate full marks.
- ii. Use nonprogrammer calculator is permitted.

ii. Use nonprogrammer calculator is permitted

Q.No.	Question	Marks	CO	BT																														
Q1	Define Project Appraisal and explain its objectives and importance in project planning.	05	CO1	BT1																														
Q2	Explain the Payback Period method for project evaluation with advantages and limitations.	05	CO2	BT2																														
Q3	A company is considering the purchase of a new machine. Two alternative machines (X and Y) have been suggested, each having an initial cost of 20,00,000 and requiring 1,00,000 as additional working capital at the end of 1st year. Earnings after taxation are expected to be as follows. <table><tr><th>Year</th><th>Machine-X</th><th>Machine-Y</th></tr><tr><td>1</td><td>2,00,000</td><td>6,00,000</td></tr><tr><td>2</td><td>6,00,000</td><td>8,00,000</td></tr><tr><td>3</td><td>8,00,000</td><td>10,00,000</td></tr><tr><td>4</td><td>12,00,000</td><td>6,00,000</td></tr><tr><td>5</td><td>8,00,000</td><td>4,00,000</td></tr></table> <table><tr><th>Year</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th></tr><tr><td>PV</td><td>0.91</td><td>0.83</td><td>0.75</td><td>0.68</td><td>0.62</td></tr></table> The company has target for a return of capital of 10% and on this basis, you are required to prepare the profitability of the machines and state which alternative you consider financially preferable using NPV method.	Year	Machine-X	Machine-Y	1	2,00,000	6,00,000	2	6,00,000	8,00,000	3	8,00,000	10,00,000	4	12,00,000	6,00,000	5	8,00,000	4,00,000	Year	1	2	3	4	5	PV	0.91	0.83	0.75	0.68	0.62	05	CO3	BT5
Year	Machine-X	Machine-Y																																
1	2,00,000	6,00,000																																
2	6,00,000	8,00,000																																
3	8,00,000	10,00,000																																
4	12,00,000	6,00,000																																
5	8,00,000	4,00,000																																
Year	1	2	3	4	5																													
PV	0.91	0.83	0.75	0.68	0.62																													
Q4	Explain the TOPSIS method used in Project selection	05	CO2	BT2																														

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO	On successful completion of this course, students should be able to:
CO 1	Appraise the selection and initiation of individual projects and its portfolios in an enterprise.
CO 2	Analyze the project planning activities that will predict project costs, time schedule, and quality.
CO 3	Develop processes for successful resource allocation, communication, and risk management.
CO 4	Evaluate effective project execution and control techniques that results in successful project completion



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of BSHM
In-Semester Examination II
Academic Year: 2025-26 (Semester II)

Class: SY

Course: Constitution of India & Governance

Course Code: MDM-IS-PC202

Date: 29.03.2026

Time: 14.30 to 15:15

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.			Marks	CO	BT
		Answer ANY TWO of the following questions. (100 Words)			
Q.1	a	Compare and contrast the features (At least 3) and respective merits and demerits (At least 1 each) of the 'Parliamentary System' and the 'Presidential System'.	5	CO2	BT2
	b	List out any FIVE key differences between 'Federal' and 'Unitary' form of governments with respective examples.	5	CO1	BT1
	c	What are the constitutional provisions with respect to different types of emergencies in India?	5	CO5	BT1
	d	Identify the special provisions for some states within the constitutional provisions from Art. 371 to 371-J.	5	CO4	BT3
		Answer ANY ONE of the following questions. (150 Words)			
Q.2	a	Examine various constitutional powers and functions of the President of India as a nominal 'Head of the State'.	10	CO1	BT4
	b	Evaluate different powers and functions of the Governor as the constitutional 'Head of the State Government'.	10	CO3	BT5

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To understand important articles of the Constitution of India & broader aspects of Governance.
CO 2	To embrace the constitutional values envisioned by the founding fathers of the nation.
CO 3	To use contextual knowledge to evaluate and address societal, health, safety, legal, and cultural issues.
CO 4	To connect syllabus topics with current national and societal issues to function effectively in multidisciplinary contexts.
CO 5	To develop the preparation and ability for independent and lifelong learning.